



浙江大学

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6.5

$$I_C = I_S e^{\frac{V_{BE}}{V_T}}$$

$$1\text{mA} = 10^{-16}\text{A} \cdot e^{\frac{V_{BE}}{25\text{mV}}} \Rightarrow V_{BE} = 748.34\text{mV}$$

$$I_{SE} = I_S / \sigma = 1.01 \times 10^{-16}\text{A}$$

$$I_{SB} = I_S / \beta = 10^{-18}\text{A}$$

6.10

$$I_E = 2\text{mA}$$

$$I_C = I_E \cdot \frac{\beta}{\beta+1} = 2\text{mA} \cdot \frac{50}{51} = 1.961\text{mA}$$

$$I_B = I_E - I_C = 2\text{mA} \cdot \frac{1}{51} = 39.216\mu\text{A}$$

$$I_C = I_S e^{\frac{V_{BE}}{V_T}}$$

$$\therefore V_{BE} = V_T \cdot \ln \frac{I_C}{I_S} = 650\text{mV}$$

6.13

$$I_E = \frac{9.3\text{V}}{10\text{k}\Omega} = 0.93\text{mA}$$

$$I_B = I_E \cdot \frac{1}{51} = 18.2\mu\text{A}$$

$$I_C = I_E \cdot \frac{50}{51} = 0.91\text{mA}$$

$$V_C = 10 - I_C \cdot 5\text{k}\Omega = 5.45\text{V}$$

6.14

$$I_B = \frac{1\text{V}}{100\text{k}\Omega} = 10\mu\text{A}$$

$$I_E = \frac{10 - 1.7}{5\text{k}} = 1.66\text{mA}$$

$$V_C = -10 + 5\text{k} \cdot 1.65\text{mA} = -1.75\text{V}$$

$$\therefore \beta+1 = 166 \Rightarrow \beta = 165, \sigma = \frac{165}{166} = 0.994 \Rightarrow I_C = 1.65\text{mA}$$



(6.17) $r_D = \frac{V_A}{I_C}$, $V_A = 100V$

当 $I_C = 0.1, 1, 10mA$ 时

$$r_D = 1M\Omega, 100k\Omega, 10k\Omega$$

(6.21) $I_B = \frac{1.7 - 0.7}{10K} = 0.1mA$

$$I_C = \beta \cdot I_B = 5mA$$

$$V_C = 10 - 1K \cdot 5mA = 5V > 0.7V \quad \therefore \text{工作在 active mode}$$

(a) at the edge of saturation, $V_C = 0.7 - 0.4 = 0.3V$

$$R_C = \frac{10 - 0.3}{5} = 1.94K\Omega$$

(b) at deep saturation, $V_C = 0.2V$

$$I_C = \beta_{forced} \times I_B = 1mA$$

$$R_C = \frac{10 - 0.2}{1mA} = 9.8K\Omega$$